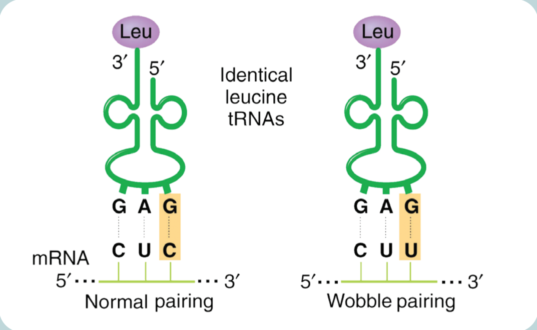
- UAA, UAG and UGA) codons are designated as termination (stop) codons and also are known as "non-sense" codons.

5.10.1 Mutation and genetic code

Comparative studies of mutations (sudden change in a gene) and corresponding alteration in amino acid sequence of specific protein have confirmed the validity of the genetic code. The relationship between genes and DNA are best understood by mutation studies. The simplest type of mutation at the molecular level is a change in nucleotide that substitutes one base for another. Such changes are known as base substitutions which may occur spontaneously or due to the action of mutagens. A well studied example is sickle cell anaemia in humans which results from a point mutation of an allele of β-haemoglobin gene (βHb). A haemoglobin molecule consists of four polypeptide chains of two types, two α chains and two β-chains. Each chain has a heme group on its surface. The heme groups are involved in the binding of oxygen. The human blood disease, sickle cell anaemia is due to abnormal haemoglobin. This abnormality in haemoglobin is due to a single base substitution at the sixth codon of the beta globin gene from GAG to GTG in β -chain of haemoglobin. It results in a change of amino acid glutamic acid to valine at the 6th position of the β -chain. This is the classical

Wobble Hypothesis

It is a hypothesis proposed by Crick (1966) which states that tRNA anticodon has the ability to wobble at its 5' end by pairing with even non-complementary base of mRNA codon. According to this hypothesis, in codon-anticodon pairing the third base may not be complementary. The third base of the codon is called wobble base and this position is called wobble position. The actual base pairing occurs at first two positions only. The importance of Wobbling hypothesis is that it reduces the number of tRNAs required for polypeptide synthesis and it overcomes the effect of code degeneracy.


Leu
3′
5′
Identical
leucine
tRNAs
Leu
3′
5′
G A G
G A G
mRNA
C U C
C U U
5′
3′
5′
3′
Normal pairing
Wobble pairing

In the above example though the codon and the anti codon do not match perfectly, yet the required amino acid is brought perfectly. This enables the economy of tRNA, GUU, GUC, GUA and GUG code for the amino acid - Valine.

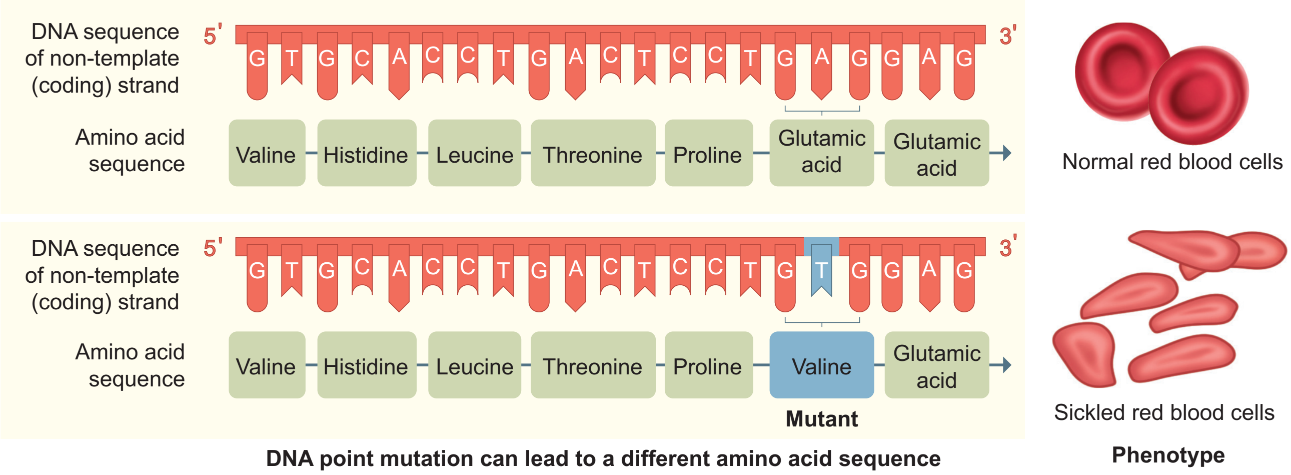

DNA sequence
of non-template
(coding) strand
5'
G T G C A C C T G A C T C C T G A G G A G
3'
Amino acid
sequence
Valine
Histidine
Leucine
Threonine
Proline
Glutamic acid
Glutamic acid
Normal red blood cells
DNA sequence
of non-template
(coding) strand
5'
G T G C A C C T G A C T C C T G T G G A G
3'
Amino acid
sequence
Valine
Histidine
Leucine
Threonine
Proline
Valine
Glutamic acid
Mutant
Sickled red blood cells
DNA point mutation can lead to a different amino acid sequence
Phenotype

Fig. 5. 10 DNA point mutation